Ivan Pena

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EDUCATION

University of Houston Expected May 2026

B.S. in Computer Science, Minor in Mathematics

Relevant Coursework: Machine Learning, Operating Systems, GPU/Parallel Computing, Data Structures & Algorithms, Computer Architecture, Interactive Computer Graphics

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript, HTML/CSS, CUDA

Frameworks & Tools: PyTorch, OpenAI API, ChromaDB, FastAPI, React.js, SQLAlchemy, Docker, Streamlit, Pandas, Git/GitHub, Linux

Concepts: LLMs, RAG, vector embeddings, AI safety & guardrails, prompt engineering, REST APIs, JWT authentication, GPU kernel optimization, parallel computing

PROJECTS

CodeAI — Full-Stack AI Code Analysis Platform *React, FastAPI, OpenAI API, ChromaDB, SQLite, Docker*

- Built a production-ready AI SaaS platform with JWT authentication, per-user dashboards, and LLM-powered code review — detecting bugs, security vulnerabilities, and performance issues as structured JSON output from GPT-3.5-turbo
- Implemented per-user RAG pipeline using OpenAI text-embedding-3-small and ChromaDB vector store, retrieving semantically similar past reviews via cosine similarity to improve feedback quality over time
- Designed RESTful API with FastAPI and SQLAlchemy ORM, containerized the full stack with Docker, and structured for cloud deployment on Render or AWS

Guardrail & Red-Team Harness *Python, FastAPI, OpenAI API, SQLite, Streamlit*

- Stress-tested LLM systems against jailbreaks, prompt injection, harmful content, and data exfiltration using a two-layer guardrail stack — regex scrubbers for known attack signatures followed by the OpenAI Moderation API for ML-based scoring
- Achieved ~92% detection accuracy with <8% false positive rate across 13 adversarial test cases; logged all attempts with severity, block reason, and per-category moderation scores to a persistent SQLite incident database
- Built a real-time Streamlit monitoring dashboard tracking block rate, category breakdowns, and false positive/negative trends across all red-team sessions

LEADERSHIP & ACTIVITIES

UH Flashcard System — Software Engineering Team Project 2024 – Present

- Drove core Java implementation of a CLI flashcard application, owning the majority of backend logic across multiple sprints in an Agile team environment
- Implemented input validation, error handling, and CLI argument parsing using the picocli framework; coordinated Git branching workflow and code reviews across the team

Code Coogs | Cougar CS — UH Computer Science Organizations 2022 – Present

- Engaged in technical workshops, collaborative coding sessions, and networking events connecting students with industry professionals in software and AI